Robust Normal-Only Industrial Anomaly Detection Under Capture Shift

A leakage-aware, calibrated, and reproducible study of detection, localization, selective review, and explanation stability on VisA

ABSTRACT

Normal-only visual anomaly detection is attractive when defective examples are rare, but benchmark accuracy can conceal brittle operating behavior. We evaluate frozen ResNet-18 multi-scale patch features, bounded k-center memory, normal-only augmentation, category-conditional conformal calibration, selective human review, and spatial evidence on the official VisA one-class split. Across 12 categories, the primary augmented 512-prototype system obtains macro image AUROC 0.8674, pixel AUROC 0.9557, and AUPRO@30 0.7762. The augmentation intervention improves shifted image AUROC by 0.0077 (95% CI [0.0030, 0.0126]). Clean normal FPR is 0.0430 with 0.8042 automatic coverage; severe blur, noise, and JPEG raise FPR to 0.8686, 0.6820, and 0.4696. Zero-shot category-transfer AUROC is 0.5783. The evidence shows that clean calibration is not a distribution-shift guarantee.

Primary takeaway: a simple bounded memory is strong and practical, but capture control and human review are part of the model boundary—not optional deployment polish.

Keywords

industrial anomaly detection · normal-only learning · robustness · conformal calibration · selective prediction · spatial evidence

1. Introduction

Industrial inspection often begins with many acceptable examples and few representative defects. Normal-only methods model acceptable visual variation and flag deviations. Their deployment risk is asymmetric: nuisance changes can create expensive false alarms, while illumination or geometry can hide defects or destabilize spatial evidence. An image AUROC on a familiar test distribution does not describe these failures.

This work asks whether a simple, auditable patch-memory system remains useful under controlled capture shifts and whether normal-only calibration can define a safer operating policy. We prioritize split integrity, false positives, category-conditional behavior, explanation stability, and reproducibility over one pooled score.

Contributions

(1) Leakage-aware VisA data governance. (2) Bounded multi-scale k-center memory. (3) A deterministic six-family shift study. (4) Category-conditional calibration and selective review. (5) A private local workstation and hash-gated public package.

2. Data governance and protocol

The audited VisA release contains 10,821 images: 7,358 training-normal, 1,301 deterministic validation-normal, and 2,162 official test images. The test set contains 1,200 anomaly and 962 normal images. Validation derives only from grouped official training-normal samples. Exact and confirmed perceptual duplicate groups may not cross splits.

Detection reports macro per-category image AUROC, average precision, and normal FPR. Localization reports pixel AUROC, pixel AP, mean IoU at a validation-normal-only threshold, and AUPRO to 30% false-positive area. Controlled shifts use two deterministic severities for JPEG, resize, blur, noise, photometric, and geometric change.

3. Method

3.1 Baselines

The sanity baseline combines colour, histogram, local binary pattern, and gradient descriptors. A global baseline uses frozen ImageNet ResNet-18 embeddings. The spatial reference retains 2,048 mid-scale projected patch prototypes per category and produces a smoothed nearest-neighbour distance map.

3.2 Bounded multi-scale memory

The proposed representation aligns frozen ResNet-18 layer-1, layer-2, and layer-3 features on a common grid. Deterministic bounded candidate sampling and greedy k-center selection compress each category to 512 prototypes. Image scoring aggregates the upper tail of the anomaly map. A balanced unified comparator uses one 1,536-entry memory.

3.3 Calibration and review

Category-conformal p-values are fit only to validation-normal scores with plus-one, tie-conservative ranks. The frozen primary policy maps $p <= 0.05$ to anomalous, $0.05 < p <= 0.20$ to review, and $p > 0.20$ to normal. Coverage is the fraction decided automatically; selective risk is the error among those decisions.

4. Clean performance

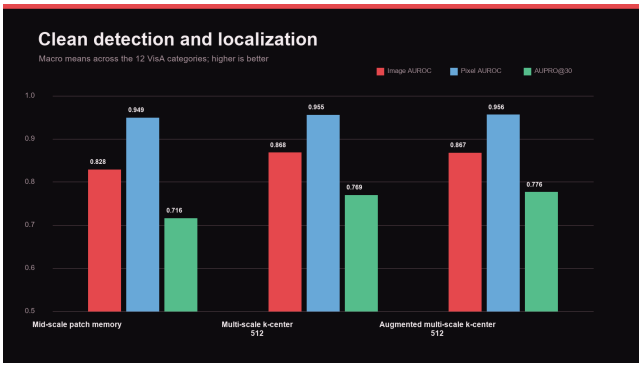


Figure 1. The multi-scale k-center memory improves ranking and AUPRO; the augmented system preserves clean performance.

System	Img AUROC	Normal FPR	Px AUROC	AUPRO
Mid-scale 2,048	0.8283	0.0477	0.9488	0.7156
Multi-scale k-center 512	0.8679	0.0471	0.9555	0.7692
Augmented k-center 512	0.8674	0.0496	0.9557	0.7762
Unified k-center 1,536	0.7936	0.0566	0.9257	0.6942

The coreset improves image AUROC by 0.0396 and AUPRO by 0.0536 over the 2,048-entry reference while cutting total category-index storage by roughly 60%. The reference retains the best frozen thresholded mean IoU (0.2507 versus 0.1665), demonstrating that threshold-free ranking and one fixed support threshold answer different questions.

5. Robustness and generalization

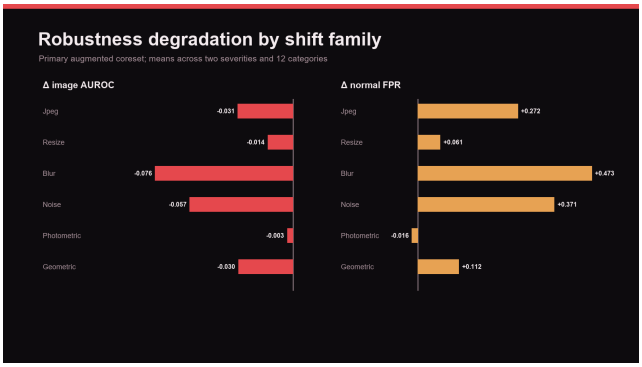


Figure 2. Blur and noise dominate detection and false-positive degradation.

Blur is the largest mean degradation: image AUROC changes by -0.0763 and normal FPR by +0.4730. Noise follows at -0.0572 and +0.3709. JPEG changes image AUROC by -0.0306 while increasing normal FPR by +0.2720. Photometric shift creates comparatively little mean AUROC change but can increase anomaly misses.

H2 is inconclusive: the multi-scale robustness advantage is -0.0072 with 95% CI [-0.0235, 0.0083]. The equal-budget augmentation intervention is supported at +0.0077 shifted image AUROC, 95% CI [0.0030, 0.0126]. Retained-correct decisions have heatmaps 0.2153 more correlated with clean evidence than failures, supporting H5.

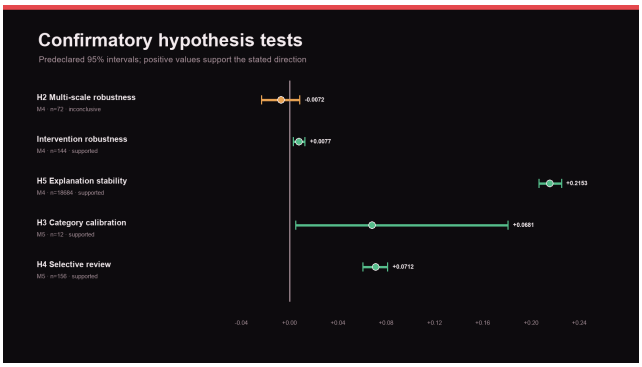


Figure 3. Four hypotheses/interventions are supported; H2 remains inconclusive.

Leave-one-category-out transfer obtains image AUROC 0.5783, 0.2896 below the category-specific system. A known category and verified category memory are therefore operating requirements.

6. Calibration and selective review

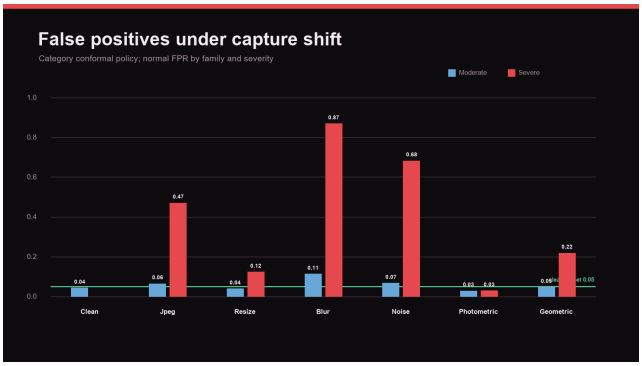


Figure 4. The clean 5% target does not transfer to severe capture shift.

Clean category-conformal normal FPR is 0.0430, review rate is 0.1958, automatic coverage is 0.8042, and selective risk is 0.1940. Category calibration improves absolute 5% false-alarm alignment over pooled calibration by 0.0681, 95% CI [0.0050, 0.1809]. The review band reduces automatic risk by 0.0713, 95% CI [0.0606, 0.0812].

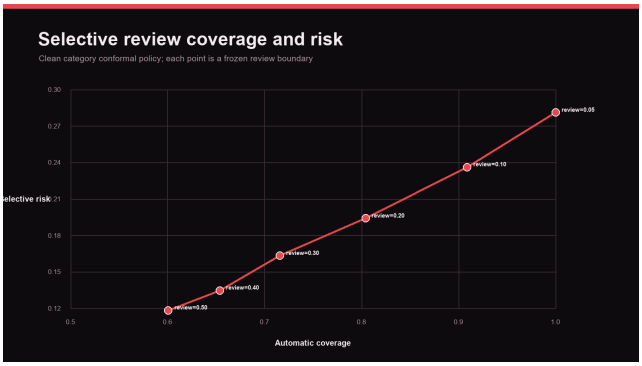


Figure 5. Wider review regions trade automatic coverage for lower automatic-decision risk.

Severe blur, noise, JPEG, and geometry produce normal FPR 0.8686, 0.6820, 0.4696, and 0.2179. Clean conformal p-values rely on exchangeability with calibration-normal scores; a changed capture distribution violates that assumption.

7. Error taxonomy

Code	Failure	Evidence / response
E1	Nuisance false positives	Severe blur adds 795 FPs. Reject degraded capture and monitor drift.
E2	Photometric misses	98 new FNs under severe photometric shift. Control illumination.
E3	Spatial instability	Geometric heatmap correlation 0.442. Evidence is not causal diagnosis.
E4	Category transfer	LOCO AUROC 0.578. Require known category memory.
E5	Calibration failure	Shift FPR exceeds clean target. Pause automatic use.
E6	Support trade-off	AUPRO improves while frozen mean IoU trails reference.

8. Practical system

The local CPU workstation loads one frozen backbone and verified indexes, accepts up to 24 images, and displays the evaluated crop, continuous heatmap, frozen support contour, score, category-conformal p-value, policy decision, thresholds, timings, and limitations. It binds to loopback, decodes in memory, and exports metadata-only reports.

Resource	Measured value
Median end-to-end latency	77.3 ms
p95 end-to-end latency	108.8 ms
Sequential throughput	12.55 images/s
Peak process working set	548.7 MB
Runtime artifacts	49.97 MB

9. Reproducibility

Every reportable run stores resolved configuration, seed, environment, Git state, manifest identity, model identity, predictions, metrics, and hashes. M7 pins the exact completed M2–M6 runs and verifies their metadata and required-file hashes before rebuilding public tables and figures. A clean isolated audit passed 54 tests, lint, strict typing, lock validation, zero-output reconstruction, and source/wheel builds.

10. Limitations and responsible use

VisA is a research benchmark, and controlled corruptions do not represent every production sensor or process. Heatmaps express feature-space difference, not physical cause. Category transfer is weak, thresholded localization is operating-point sensitive, and CPU measurements come from one workstation. No factory trial or certified quality guarantee is claimed.

Use requires a known category, verified memory and calibration artifacts, controlled acquisition, visible human review, and shift monitoring. A capture change should pause automatic use until corrected or revalidated with approved normal data.

11. Conclusion

A bounded multi-scale k-center memory is a strong clean baseline, and equal-budget normal augmentation produces a supported robustness gain. Category calibration and selective review improve clean operating behavior. The dominant boundary is equally clear: shift can overwhelm false-positive control, spatial evidence can destabilize, and unseen-category transfer is unreliable. Reporting these limits alongside accuracy is the central contribution.

References

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